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Name: _____

May 13, 2018 Sunday

Department/School: _____

SID: _____

University Physics I

Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1. (20pts.) 1.(20pts.) A boat of mass m is pushed from the shore into a lake with initial velocity v_0 . The motion of the boat is resisted by a force $F = -mkv$ where v is the velocity of the boat and k is a constant.

(a) Find the velocity of the boat as a function of time.

(b) Find the velocity of the boat as a function of the distance from the shore.

(c) What is the total distance that the boat will travel?

Solution: (a) The equation of motion is

$$m \frac{dv}{dt} = -mkv.$$

Separation of variable gives

$$\frac{dv}{v} = -kdt.$$

Integrating both sides of above equation, we obtain

$$\int_{v_0}^v \frac{dv}{v} = -k \int_0^t dt \quad \text{or} \quad \ln \frac{v}{v_0} = -kt \quad \text{or} \quad v = v_0 e^{-kt}.$$

(b) The equation of motion can be rewritten as

$$m \frac{dv}{dx} \frac{dx}{dt} = mv \frac{dv}{dx} = -mkv \quad \text{or} \quad \frac{dv}{dx} = -k \quad \text{or} \quad dv = -kdx.$$

Integrating both sides of above equation, we obtain

$$\int_{v_0}^v dv = -k \int_0^x dx \quad \text{or} \quad v = v_0 - kx.$$

(c) When $v = v_0 - kx = 0$, we have $x = \frac{v_0}{k}$. That is the total distance the boat travels.

2. (20pts.) The so-called double-well potential can be modeled as the one-dimensional potential energy function

$$U(x) = -\frac{1}{2}\alpha x^2 + \frac{1}{4}\beta x^4$$

where α and β are both positive constants.

(a) Sketch the potential energy diagram. Draw in the diagram the kinetic energy of the particle at a position.

(b) Find the corresponding force acting on the particle.

(c) Find the equilibrium positions of the particle and identify them as stable or unstable.

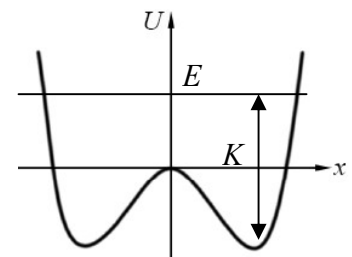
Solution: (a)

$$(b) F(x) = -\frac{dU}{dx} = \alpha x - \beta x^3$$

(c) From $F(x) = \alpha x - \beta x^3 = 0$, we have three equilibrium positions at $x = 0$ and

$$x = \pm \sqrt{\frac{\alpha}{\beta}}.$$

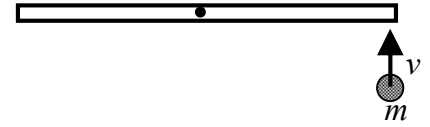
The second order derivative is $\frac{d^2U}{dx^2} = -\alpha + 3\beta x^2$. Thus



at $x = 0$, $\frac{d^2U}{dx^2} = -\alpha < 0$, it is unstable;

at $x = \pm \sqrt{\frac{\alpha}{\beta}}$, $\frac{d^2U}{dx^2} = 2\alpha > 0$, they are stable.

3. (20pts.) A uniform rod of length L and mass M is pivoted without friction at its center so that it can rotate freely in the vertical plane. Initially the rod is at rest horizontally. A wad of putty of mass m hits the right end of the rod with a vertically upward velocity v and then sticks to the rod.



(a) What is the angular speed of the rod after the hitting?

(b) What is the ratio of the total kinetic energies just after and before the hitting?

(c) Through what angle will the rod rotate before it momentarily stops?

Solution: (a) The conservation of angular momentum gives

$$mv \frac{L}{2} = \left[\frac{1}{12} ML^2 + m \left(\frac{L}{2} \right)^2 \right] \omega.$$

Thus

$$\omega = \frac{2mv}{\left(\frac{M}{3} + m \right) L}.$$

(b) Before hitting: $K_i = \frac{1}{2} mv^2$.

$$\text{After hitting: } K_f = \frac{1}{2} I \omega^2 = \frac{1}{2} \left[\frac{1}{12} ML^2 + m \left(\frac{L}{2} \right)^2 \right] \left[\frac{2mv}{\left(\frac{M}{3} + m \right) L} \right]^2 = \frac{1}{2} \frac{m^2 v^2}{\frac{M}{3} + m}.$$

The ratio: $\alpha = \frac{K_f}{K_i} = \frac{m}{\frac{M}{3} + m}$.

(c) The conservation of mechanical energy gives

$$K_f = \frac{1}{2} \frac{m^2 v^2}{\frac{M}{3} + m} = mg \frac{L}{2} \sin \theta.$$

$$\text{Thus } \sin \theta = \frac{mv^2}{\left(\frac{M}{3} + m \right) gL} \quad \text{or} \quad \theta = \sin^{-1} \left[\frac{mv^2}{\left(\frac{M}{3} + m \right) gL} \right].$$

4. (20pts.) A satellite is moving uniformly in a circular orbit of radius $4R_E$ about the centre of the earth, where R_E is the radius of the earth. Retro-rockets are fired to reduce the speed of the satellite from v to αv ($0 < \alpha < 1$), without change in direction, so that it subsequently moves in an elliptic orbit and the least distance from the center of the earth is $2R_E$. Find the value of α .

Solution: The speed of the satellite in circular motion is determined by

$$G \frac{M_E m}{(4R_E)^2} = m \frac{v^2}{4R_E}.$$

Thus $v = \frac{1}{2} \sqrt{\frac{GM_E}{R_E}}$. The speed is reduced to αv and the energy of subsequent motion is

$$E = \frac{1}{2} m \alpha^2 v^2 - \frac{GM_E m}{4R_E} = \frac{GM_E m}{8R_E} (\alpha^2 - 2).$$

The conservation of angular momentum gives

$$m(4R_E)\alpha v = m(2R_E)v_p, \quad v_p = 2\alpha v = \alpha \sqrt{\frac{GM_E}{R_E}}.$$

where v_p is the speed of the satellite at the perigee. The conservation of energy gives

$$\frac{1}{2}mv_p^2 - \frac{GM_E m}{2R_E} = \frac{GM_E m}{2R_E}(\alpha^2 - 1) = \frac{GM_E m}{8R_E}(\alpha^2 - 2).$$

Thus

$$4(\alpha^2 - 1) = (\alpha^2 - 2), \quad 3\alpha^2 = 2, \quad \alpha = \sqrt{\frac{2}{3}}.$$

5. (20 pts.) A uniform cylinder of mass m and radius R has a light inextensible string wound around it. The cylinder is placed on an inclined frictionless surface that makes an angle α to the horizontal. The free end of the string is attached to the wall as shown in the figure.

(a) Write the equation of translational motion of the center of mass and the equation of the rotational motion of the cylinder.

(b) Find the center-of-mass acceleration and the rotational acceleration of the cylinder.

(c) If the cylinder is released from rest and it descends the inclined surface of length L , what are the center-of-mass velocity and the angular velocity of the cylinder?

Solution: (a) The equation of translational motion is

$$m \frac{dv}{dt} = mg \sin \alpha - T. \quad (1)$$

The equation of rotational motion is

$$I \frac{d\omega}{dt} = TR, \quad I = \frac{1}{2}mR^2. \quad (2)$$

The condition for rolling without slipping is

$$v = \omega R \quad \text{or} \quad \omega = \frac{v}{R}. \quad (3)$$

(b) Substituting $I = \frac{1}{2}mR^2$ and $\omega = \frac{v}{R}$ into Eq. (2), we have

$$T = \frac{1}{2}m \frac{dv}{dt}. \quad (4)$$

Substituting Eq. (4) into Eq. (1), we have

$$m \frac{dv}{dt} = mg \sin \alpha - \frac{1}{2}m \frac{dv}{dt}. \quad (5)$$

Therefore $\frac{3}{2}m \frac{dv}{dt} = mg \sin \alpha$ and

$$a_c = \frac{dv}{dt} = \frac{2}{3}g \sin \alpha,$$

$$\frac{d\omega}{dt} = \frac{1}{R} \frac{dv}{dt} = \frac{2g \sin \alpha}{3R}.$$

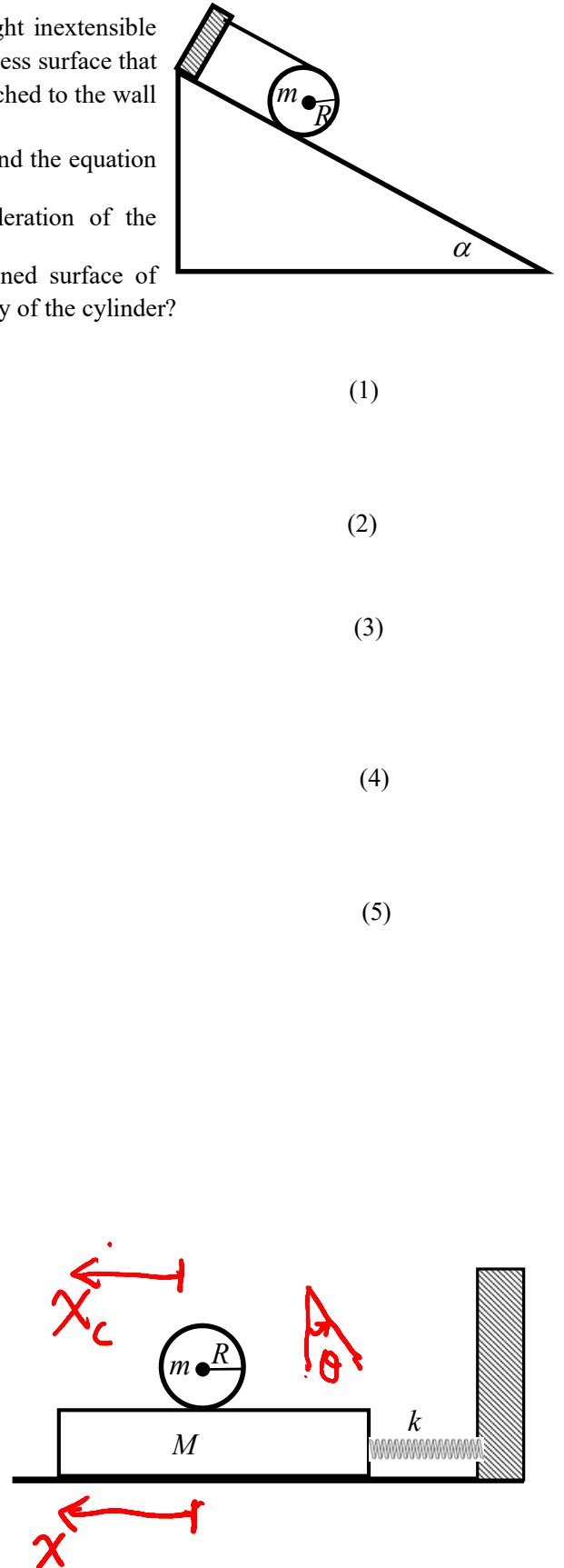
(c) $v = \sqrt{2a_c L} = \frac{2}{3}\sqrt{3gL \sin \alpha}, \quad \omega = \frac{v}{R} = \frac{2}{3R}\sqrt{3gL \sin \alpha}.$

6. (20pts.) A block of mass M moves on a frictionless horizontal surface. The block is connected to the wall with a spring of spring constant k . A uniform sphere of mass m and radius R is placed on the horizontal top of the block and it rolls without slipping on it.

(a) Write the equation of translational motion of the block.

(b) Write the equations of translational and rotational motions of the ball.

(c) Find the (angular) frequency of the oscillation of the block.



Solution: (a) The equation of translational motion of the block is

$$M\ddot{x} = -kx + F_f . \quad (1)$$

(b) The equation of translational motion of the ball is

$$m\ddot{x}_C = -F_f . \quad (2)$$

The equation of rotational motion of the ball is

$$I\ddot{\theta} = F_f R, \quad I = \frac{2}{5}mR^2 . \quad (3)$$

The condition for the ball to roll without slipping on the top of the block gives

$$R\ddot{\theta} = \ddot{x}_C - \ddot{x} . \quad (4)$$

(c) Substituting (2) into (1) gives

$$M\ddot{x} + m\ddot{x}_C = -kx . \quad (5)$$

Substituting $\ddot{x}_C = \ddot{x} + R\ddot{\theta}$ into (5) gives

$$(M + m)\ddot{x} + mR\ddot{\theta} = -kx . \quad (6)$$

Substituting (2) into (3), we obtain

$$\frac{2}{5}mR^2\ddot{\theta} = -mR\ddot{x}_C = -mR\ddot{x} - mR^2\ddot{\theta} \quad \text{or} \quad mR\ddot{\theta} = -\frac{5}{7}m\ddot{x} . \quad (7)$$

Substituting (7) into (6), we have

$$\left(M + \frac{2m}{7}\right)\ddot{x} = -kx \quad \text{or} \quad \ddot{x} + \frac{k}{M + \frac{2m}{7}}x = 0 .$$

Therefore, $\omega = \sqrt{\frac{k}{M + \frac{2m}{7}}} .$